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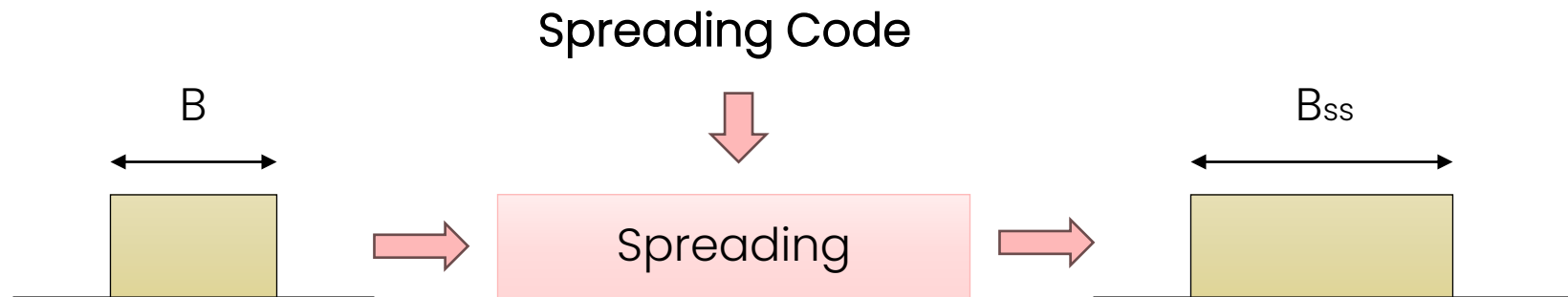
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SPREAD SPECTRUM

- ❑ Spread Spectrum is a wireless communication technology that distributes the transmitted signal across a larger bandwidth than the original signal. This broader bandwidth provides:
- Resistance to noise
 - Improved security and privacy
 - Multiple user access
 - High noise immunity
 - Immune to jamming



- ❑ There are two techniques used in spread spectrum: 1. Frequency hopping spread spectrum (FHSS) and 2. Direct sequence spread spectrum (DSSS)

PSEUDO NOISE SEQUENCE

- ❑ A coded sequence of 1s and 0s with certain auto-correlation properties, called as **Pseudo-Noise Code** is used in spread spectrum techniques.
- ❑ It is a deterministic binary sequence that appears random but is actually generated by a known algorithm and can be reproduced exactly at the receiver.
- ❑ It helps detect error in transmitted signal over channel/medium at receiver end.

- Spread spectrum method is used in Code Division Multiple Access (CDMA).
- In Frequency division multiple access (FDMA) and Time division multiple access (TDMA) the bandwidth is limited , low spectral efficiency, concentrated spectrum & power, narrow band easily affected by noise.
- This is overcome by spread spectrum.
- Spread spectrum has interference immunity, low power density, high bit rate, security due to larger bandwidth.

FREQUENCY HOPPING SPREAD SPECTRUM

- ❑ In **FHSS** technique, the carrier frequency rapidly switches among many frequency channels according to a pseudo-random sequence known to both transmitter and receiver.
- ❑ The pattern is determined by pseudo noise code.
- ❑ For example, a frequency was allotted to sender 1 for a particular period of time. After a while, sender 1 hops to the other frequency and sender 2 uses the first frequency. This is called as **frequency reuse**.
- ❑ The amount of time spent on each frequency hop is called as **Dwell time**.

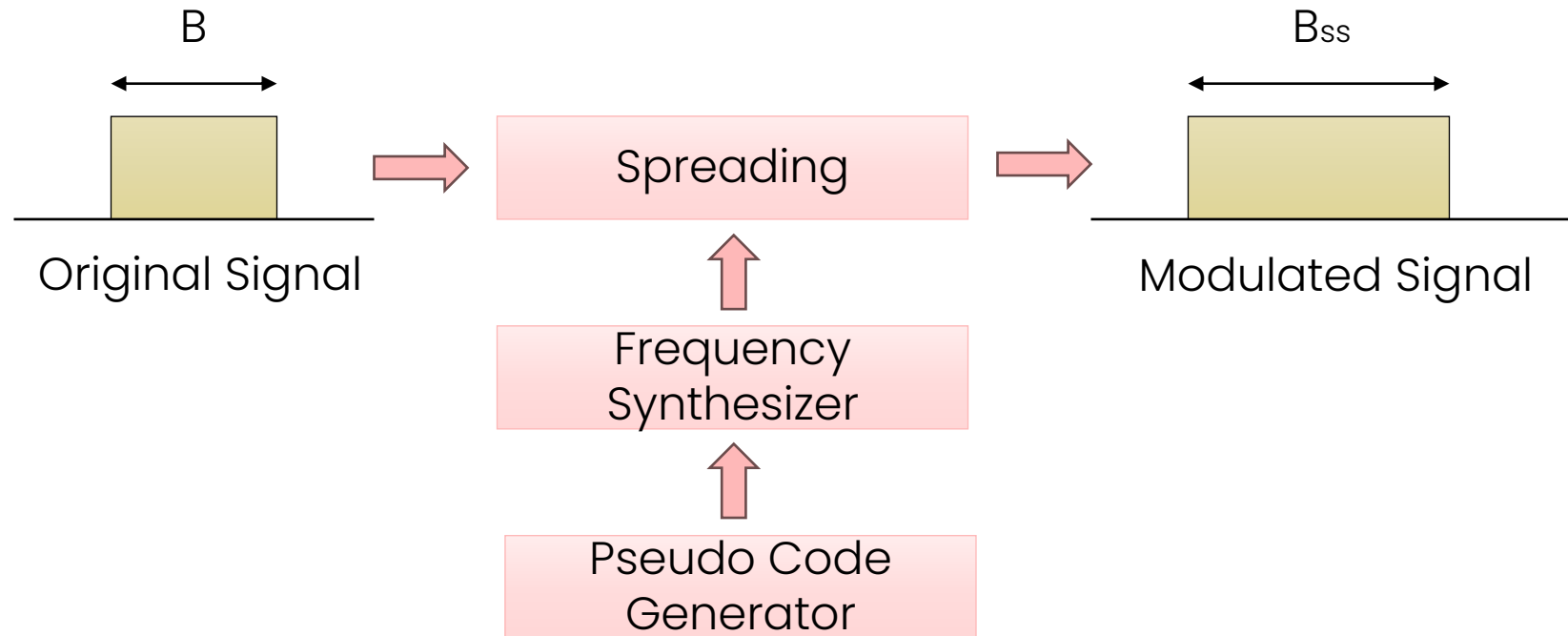
Advantages:

- Sender doesn't need to wait
- Signal has high strength
- Never affected by interference
- This technique is commonly used and affordable
- Processing Gain is higher than DSSS

Disadvantages:

- Hard to find user frequency at given time
- Requires precise synchronization
- Complex frequency control

FREQUENCY HOPPING SPREAD SPECTRUM



❑ **Applications:** Bluetooth, military communication, walkie-talkie, wireless local area network

DIRECT SEQUENCE SPREAD SPECTRUM

- ❑ DSSS is a spread spectrum technique in which the data signal is multiplied by a high-rate pseudo-noise code to spread it over a much wider frequency band.
- ❑ Each and every bit of the user data is multiplied by a secret code, called as **chipping code**. It also helps retrieve the original message at receiver end.

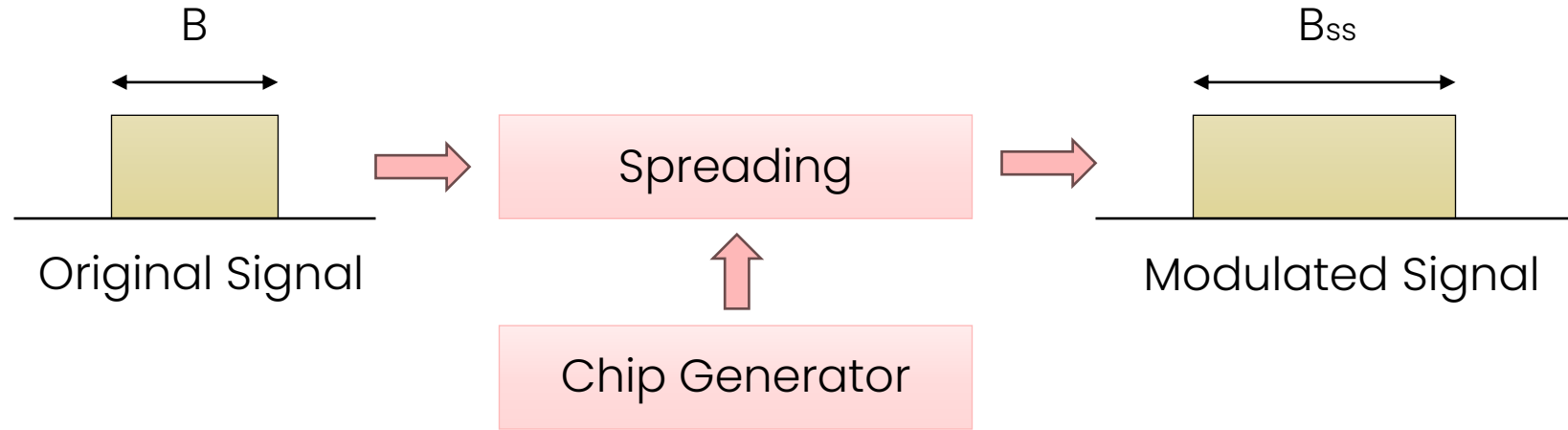
Advantages:

- Most effective solution to jamming
- DSSS performance in presence of noise is superior to FHSS.
- Minimizes signal interference

Disadvantages:

- Signal power strength is low
- Expensive compared to FHSS
- Complex receiver

DIRECT SEQUENCE SPREAD SPECTRUM



❑ **Applications:** Global positioning system (GPS), cellular network, satellite communication



THANK YOU

